

Business Requirements Document

AstraCart Ecommerce Platform

AstraCart is a hypothetical mid-market ecommerce platform intended to validate end-to-end ingestion and generation quality. The platform supports customer account access, catalog discovery, checkout, payment gateway integration, payment result handling, and post-order tracking.



Business goals: increase successful checkout conversion, reduce support contacts for payment uncertainty, expose complete order tracking, and support marketing-driven discovery without weakening security or accessibility.

Business KPI	Baseline	Target	Measurement
Registration completion	62%	78%	Analytics funnel AUTH_REG_COMPLETE
Login recovery success	44%	70%	Password reset token completion
Checkout conversion	31%	42%	Checkout start to paid order
Payment uncertainty tickets	11k/month	-35%	Support reason PAYMENT_UNKNOWN
Order tracking self-service	48%	75%	Track page views before ticket

Scope and Personas

Persona	Primary Need	Critical Journey	Risk
Guest shopper	Browse and checkout quickly	Home -> PDP -> Checkout	
Registered customer		Login -> Orders -> Track	Session expiry during payment
Support associate	Resolve payment/order ambiguity	Search order -> payment event trail	Incorrect refund guidance
Merchandiser	Manage home/catalog campaigns	Campaign setup -> category rail	Wrong segment targeting
Finance ops	Reconcile payment settlement	Gateway webhook -> order status	Duplicate capture/refund



Customer Identity: Registration, login, logout, password recovery, risk-based verification, account profile.

Shopping Discovery: Home page, search, product listing, filters, product detail, recommendations.

Checkout: Cart review, address, shipping promise, coupon, tax, payment option selection.

Payments: Gateway order creation, redirect/intent, success, failure, pending, refund triggers.

Orders: Order history, tracking, invoice, cancellation, return/refund eligibility.

Module Requirements Matrix

Req ID	Module	Requirement	Priority	Acceptance Signal
BR-AUTH-01	Registration	Email/mobile registration with duplicate-account detection	Must	
BR-AUTH-07	Reset	Reset link expires and can be resent with throttling	Must	Expired token returns safe messa
BR-CAT-02	Listing	Filters preserve URL state and inventory visibility	Must	Filter reload keeps same result se
BR-PDP-04	Detail	Variant and delivery promise gate Add to Cart	Should	Invalid SKU blocks cart action
BR-CHK-06	Checkout	Price lock survives retry within configured window	Must	Retry retains order quote id
BR-PAY-03	Payment	Gateway webhook reconciles redirect mismatch	Must	
BR-ORD-05	Orders	Order timeline exposes shipment/refund states	Must	

AstraCart

HomeCatalogCartOrdersCheckout

Checkout and Payment Gateway

Checkout combines cart review, address selection, shipment promise, offer application, tax calculation, and gateway handoff.

Cart Review

Quantity edit, promo validation, item-level stock recheck, and price lock.

QA note: verify copy, state, accessibility, and API mapping.

Address

Saved address, new address form, serviceability, and phone validation.

QA note: verify copy, state, accessibility, and API mapping.

Payment

Cards, UPI, wallets, net banking, COD, and gift card split payment.

QA note: verify copy, state, accessibility, and API mapping.

Gateway

Create payment order, redirect/intent, callback, webhook reconciliation.

QA note: verify copy, state, accessibility, and API mapping.

Fraud

Risk score, velocity checks, high-value order review.

QA note: verify copy, state, accessibility, and API mapping.

Accessibility

Keyboard focus and error summary on payment failure.

QA note: verify copy, state, accessibility, and API mapping.